

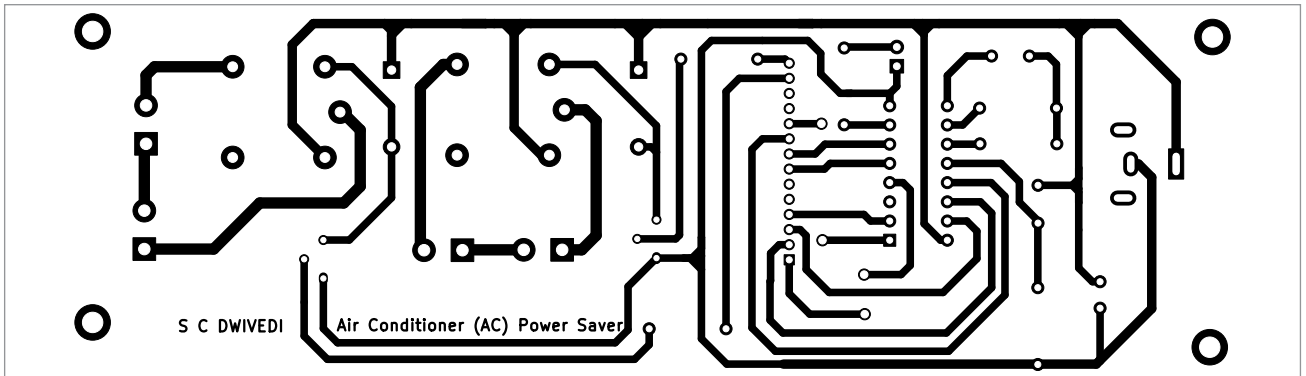


Fig. 3: Actual-size PCB

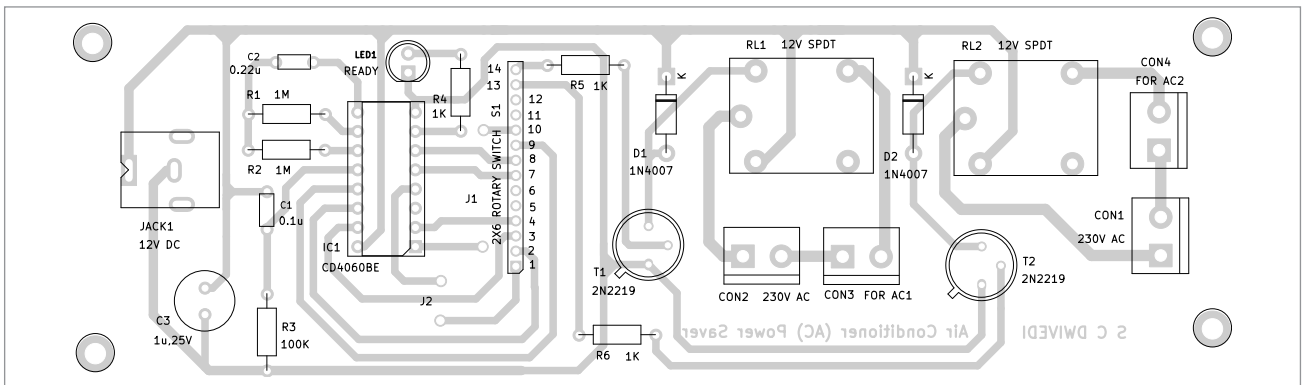


Fig. 4: Components side of the PCB

pin 13 to pin 3 (Q10). In position 4, pin 14 connects to pin 10 (Q12), and pin 13 to pin 4 (Q13). This

arrangement enables the selection of different timing output pairs from the CD4060 IC to control external

loads or relay circuits.

The circuit operates as follows: For example, if the requirement is for the first AC to be on for approximately 17 minutes and off for 17 minutes, and the second AC to be on for around 34 minutes and off for 34 minutes, then switch S1 should

**TABLE 1: OUTPUT TIMING OF CD4060**

| Positions of S1                | CD4060 Output | On Time  | Off Time | Pin14/Pin13              | Status of AC              |
|--------------------------------|---------------|----------|----------|--------------------------|---------------------------|
|                                | Q4            | 4 sec    | 4 sec    | —                        | —                         |
| Position1<br>Keep pole14 at 7  | Q5            | 8 sec    | 8 sec    | Pin 14 connects with Q5  | AC1 on/off for 8 seconds  |
| Position1<br>Keep pole13 at 1  | Q6            | 16 sec   | 16 sec   | Pin13 connects with Q6   | AC2 on/off for 16 seconds |
| Position2<br>Keep pole14 at 8  | Q7            | 32 sec   | 32 sec   | Pin 14 connects with Q7  | AC1 on/off for 32 seconds |
| Position2<br>Keep pole13 at 2  | Q8            | 64 sec   | 64 sec   | Pin13 connects with Q8   | AC2 on/off for 8 seconds  |
| Position3<br>Keep pole14 at 9  | Q9            | 128 sec  | 128 sec  | Pin 14 connects with Q9  | AC1 on/off for 8 seconds  |
| Position3<br>Keep pole13 at 3  | Q10           | 256 sec  | 256 sec  | Pin13 connects with Q10  | AC2 on/off for 8 seconds  |
| Position4<br>Keep pole14 at 10 | Q12           | 1024 sec | 1024 sec | Pin 14 connects with Q12 | AC1 on/off for 8 seconds  |
| Position4<br>Keep pole13 at 4  | Q13           | 2048 sec | 2048 sec | Pin13 connects with 14   | AC2 on/off for 8 seconds  |

**Note:** The on/off time can be adjusted by changing R1, R2, and C2.

**PARTS LIST***Semiconductors:*

- IC1 - CD4060BE
- T1, T2 - 2N2219 NPN transistor
- D1, D2 - 1N4007 rectifier diode
- LED1 - 5mm LED

*Resistors (all 1/4-watt,  $\pm 5\%$  carbon):*

- R1, R2 - 1M $\Omega$
- R3 - 100k $\Omega$
- R4-R6 - 1k $\Omega$

*Capacitors:*

- C1 - 0.1 $\mu$ F ceramic disc
- C2 - 0.22 $\mu$ F ceramic disc
- C3 - 1 $\mu$ F, 25V electrolytic

*Miscellaneous:*

- JACK1 - DC connector
- BATT. - 12V battery/12V adaptor
- RL1, RL2 - 12V SPDT relay
- S1 - 2x6 rotary switch